

SAAD FAROOQ

SECURITY RESEARCHER | ENGINEER

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PROFESSIONAL SUMMARY

Cyber Security Researcher and Security Software Engineer with expertise in applied cybersecurity research, secure software engineering, and the design of resilient, cloud-native systems. My work combines academic research with practical software engineering to develop secure applications, AI-driven cybersecurity solutions, and intelligent threat detection systems. I have experience designing and implementing secure backend platforms, distributed systems, and cybersecurity solutions using Golang, Python, cloud technologies, machine learning, and modern software architectures. My research interests include cyber threat intelligence, artificial intelligence for cybersecurity, application security, and secure system design, with a strong focus on translating research into practical solutions. Alongside my industry experience, I have taught Computer Science at college level and am passionate about delivering project-based learning, conducting applied research, and preparing the next generation of cybersecurity professionals.

EXPERIENCE

Security Software Engineer, Sheffield Hallam University

Nov 2025 to Present

Sheffield, United Kingdom

- Working on a real-time IoT healthcare platform focused on secure medical data ingestion, regulatory compliance, and cloud-native architecture
- Architected and developed secure RESTful APIs in Golang for IoT device data ingestion and analytics dashboards
- Built real-time communication systems using WebSockets and Server-Sent Events for live medical monitoring Integrated IoT gateways and devices with Google Firestore using structured validation and secure data pipelines
- Designed and deployed serverless backend infrastructure using Google Cloud Functions Implemented Redis for secure caching, performance optimization, and API rate-limiting Configured and hardened NGINX as a reverse proxy with SSL/TLS enforcement and request filtering Implemented secure authentication and authorization using OAuth 2.0 and role-based access control
- Built secure CI/CD pipelines using GitHub Actions, Golang, and Python with automated testing and security checks
- Applied OWASP Top 10 security principles to prevent injection attacks, broken authentication, and API misconfigurations
- Ensured HIPAA-aligned data protection standards for handling sensitive healthcare data Combined backend engineering, cloud security, DevOps automation, and healthcare compliance in a production-grade IoT system

Computer Science Lecturer, Punjab College

Jun 2024 to Feb 2025

Gujrat, Pakistan

- Delivered lectures in Computer Science to college-level students, covering core computing concepts, programming fundamentals, databases, and networking.
- Designed lesson plans, practical laboratory sessions, and assessments to support student learning and academic achievement.
- Guided students through practical programming exercises and problem-solving activities using an interactive, student-centred teaching approach.
- Assessed assignments, examinations, and coursework, providing constructive feedback to enhance students' technical understanding and performance.
- Fostered an engaging and inclusive classroom environment that encouraged critical thinking, collaboration, and the practical application of computing concepts.

Research Assistant , University of Management and Technology

Jan 2024 to Aug 2025

Lahore

- Conducted undergraduate research on the design and evaluation of a real-time DDoS Intrusion Detection System (IDS) using machine learning and distributed data processing.
- Designed and implemented a scalable IDS architecture using Apache Kafka, Apache Spark, Golang, Rust, and Python for real-time network traffic analysis.
- Trained and evaluated machine learning models on the CICDDoS2019 dataset, achieving up to 95.2% detection accuracy.

- Developed real-time data processing pipelines, automated mitigation mechanisms, and a web-based monitoring dashboard for intrusion detection and visualisation.
- Led the technical design, implementation, and experimental evaluation of the proposed framework, resulting in a peer-reviewed publication in IEEE Access (2026).

Backend Developer, Kloak Systems

Feb 2025 to Sept 2025

London, United Kingdom

- Designed and developed secure backend services in Golang for an NFC-enabled digital identity and event management platform.
- Engineered high-performance RESTful APIs for generating and managing Apple Wallet and Google Wallet digital passes.
- Built scalable backend systems supporting NFC-based access to event tickets, locker assignments, and digital credentials.
- Developed role-based administration dashboards with secure authentication and access control for platform management.
- Implemented PostgreSQL, Apache Kafka, and Redis to deliver secure, scalable, and high-performance data processing.
- Integrated Stripe payment services and AI-powered analytics using Large Language Model (LLM) APIs to enhance platform functionality and operational insights.
- Applied secure software engineering principles to deliver a reliable, scalable, and production-ready cloud-native solution.

EDUCATION

Sheffield Hallam University, Sheffield, United Kingdom

Sept 2025 to Present

Master of Science in Cyber Security

University of Management and Technology, Lahore, Pakistan

Sept 2020 to Aug 2024

Bachelor of Science in Computer Science

PROJECTS

DDoS Attacks Detection and Prevention on Real-Time Streaming Data Using Machine Learning Methods With Kafka and Apache Spark

Jan 2024 to Jul 2025

University of Management and Technology

- Led the design and implementation of a real-time intrusion detection and prevention framework for identifying DDoS attacks using machine learning and distributed stream-processing technologies.
- Developed a scalable architecture integrating Apache Kafka for continuous network-traffic ingestion and Apache Spark for distributed preprocessing, model training and inference.
- Prepared and analysed the CICDDoS2019 benchmark dataset, including data cleaning, feature selection and extraction of flow-based network characteristics for attack classification.
- Evaluated multiple machine-learning algorithms, including XGBoost, Random Forest, Support Vector Machine, K-Means, Naive Bayes and Logistic Regression, using accuracy, precision, recall and F1-score.
- Achieved the highest detection performance with XGBoost, reaching approximately 95.2% accuracy, followed by Random Forest and SVM.
- Designed a risk-based response mechanism that classified network events by threat level and supported mitigation actions including traffic filtering, monitoring and connection blocking.
- Conducted scalability and performance analysis to assess the suitability of the architecture for high-volume, real-time network environments.
- Led the technical development and experimental work underpinning a peer-reviewed article published in *IEEE Access*.

AI-Driven Dark Web Threat Intelligence Classification using Machine Learning and Large Language Models

Jan 2026 to Present

Sheffield Hallam University

- Conducting research into an intelligent threat intelligence framework that combines Machine Learning (ML) and Large Language Models (LLMs) to improve the classification and analysis of dark web cyber threat intelligence.
- Designing an uncertainty-gated hybrid architecture that dynamically routes complex or low-confidence samples from ML models to LLMs, improving classification accuracy while reducing computational cost.
- Evaluating multiple approaches using publicly available dark web datasets and benchmarking performance based on accuracy, latency, reliability, cost, and calibration metrics.

- Applying data preprocessing, feature engineering, TF-IDF vectorisation, and model calibration techniques to enhance threat classification performance.
- Investigating the practical application of AI-driven threat intelligence to support cybersecurity analysts in identifying emerging threats and improving security decision-making.
- Research findings will contribute to the understanding of scalable and cost-effective AI-assisted cyber threat intelligence systems.

SKILLS

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| • Application Security | • Golang | • Project Management | • Machine Learning |
| • Threat Modelling | • Python | • Technical Report Writing | • Project Supervision |
| • Secure Code Review | • AWS | • Literature Review | • Performance Evaluation |
| • Vulnerability Assessment | • Nginx | • Experimental Design | • Scientific Writing |
| • Security Research | • Kafka | • Literature Review | |

LINKS

[GitHub](#), [Linkedin](#)

AWARDS

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| Cyber Hackathon Pakistan Winner , Government of Pakistan | Jun 2023 |
| Student Researcher of the Year , University of Management and Technology | Aug 2024 |

PUBLICATIONS

Ahmad, J., Ahmed, U., Farooq, S., Sarwar, M., Shaheen, M., Manias, D. M., & Khan, J. (2026). *DDoS Attacks Detection and Prevention on Real-Time Streaming Data Using Machine Learning Methods With Kafka and Apache Spark*. **IEEE Access**, 14, 84163–84175. <https://doi.org/10.1109/ACCESS.2026.3696041>